PATIENT CARE THEORY 2

UNIT 8: Part 2 – AKI and CKD

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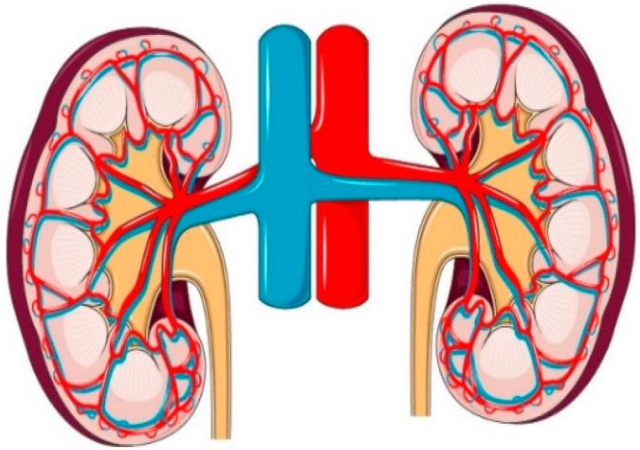
Acute Kidney Injury

- Formerly known as Acute renal failure (ARF)
- Toxic build up of waste products impair kidney function
- defined as a precipitous and significant (>50%) decrease in glomerular filtration rate (GFR)
  - over a period of hours to days
  - with an accompanying accumulation of nitrogenous wastes (e.g. creatinine, blood urea nitrogen [BUN],) in the body

Risk Factors/ Causes

Susceptibilities

- Elderly
- Diabetes
- Chronic kidney disease
- Dehydration or volume depletion
- Chronic diseases (heart, lung, liver)
- Female gender
- Cancer
- Anemia



Exposures

- Sepsis
- Critical illness
- Circulatory shock
- Burns
- Trauma
- Cardiac surgery
- Major noncardiac surgery
- Nephrotoxic drugs
- Radiocontrast agents

Main pathogenesis of AKI

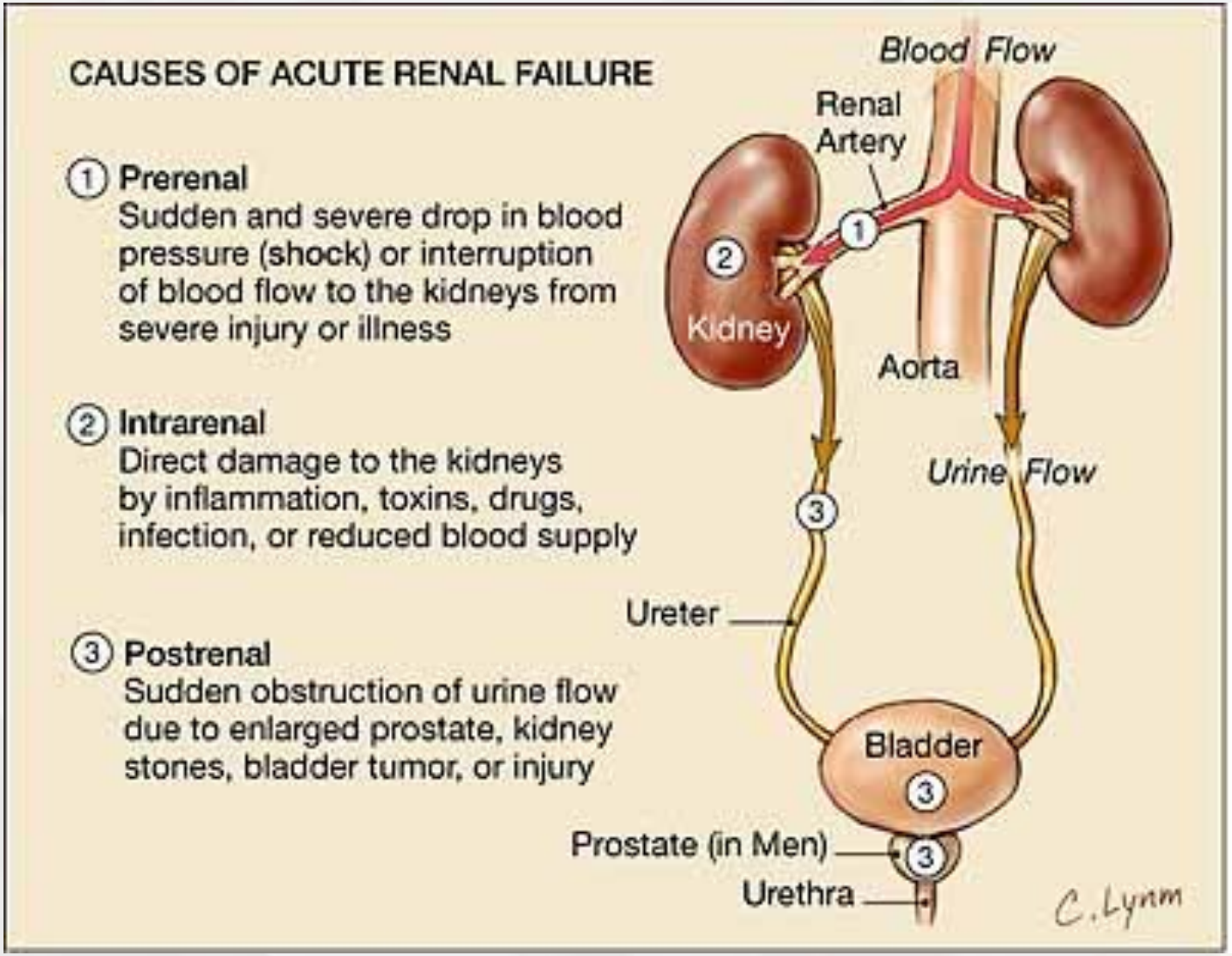
- Decreased renal perfusion
- Hypoxic injury
- Endothelial dysfunction
- Oxidative stress
- Inflammatory infiltration
- Inappropriate complement activation
- Formation of microthrombi
- Cytokine-induced injury
- Direct tubular injury
- Obstruction

Acute Kidney Injury

- Etiologies
  - reductions in renal blood flow (RBF) represent a common pathologic pathway for decreasing GFR
  - Pre-renal
  - Intrarenal
  - Post-renal

GFR

Acute Kidney Injury - Causes



Acute Kidney Injury

- It is on the rise
- incidence of community-acquired AKI (presents on admission to hospital) is as high as 25% of the population in some areas
- hospital acquired AKI is ~ 10 - 15% of patients admitted to hospital and more than 60% of those admitted to the ICU (Acute Kidney Injury - StatPearls - NCBI Bookshelf (nih.gov))

What does AKI look like?

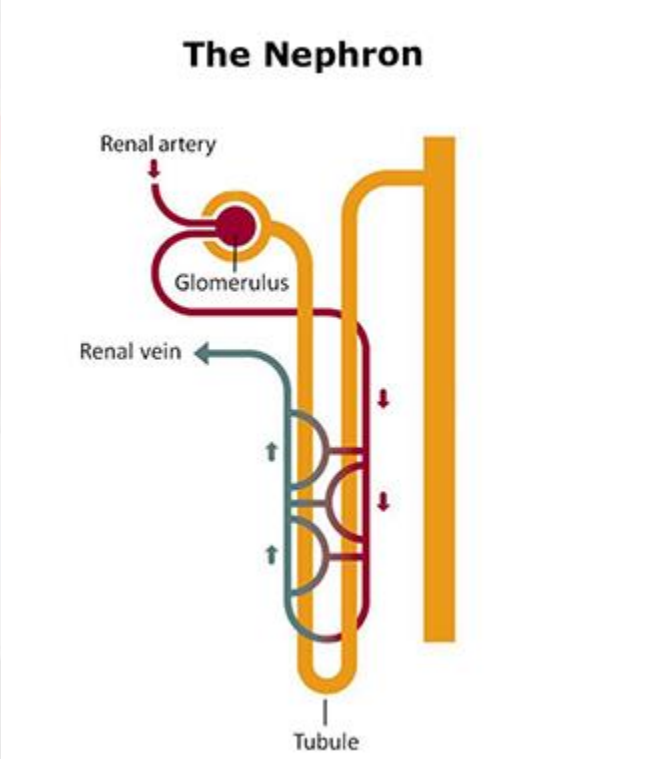
- Decreased urine output
- Edema to the legs, ankles, face, hands
- Fatigue
- SOB
- Confusion/ Altered LOA
- Nausea

- In some cases there are no symptoms, it is found through blood work and urinalysis
- Patients at risk and showing signs require in hospital treatment

Chronic Kidney Disease

- Formerly called Chronic Renal Failure (CRF)
- Characterized by inadequate kidney function due to permanent loss of nephrons
  - Clinical signs develop at a 70% loss of nephrons
  - 80% = end stage renal failure; will require dialysis and/or kidney transplant
- Combination of DM and HTN more than half of all cases of CKD
- High mortality

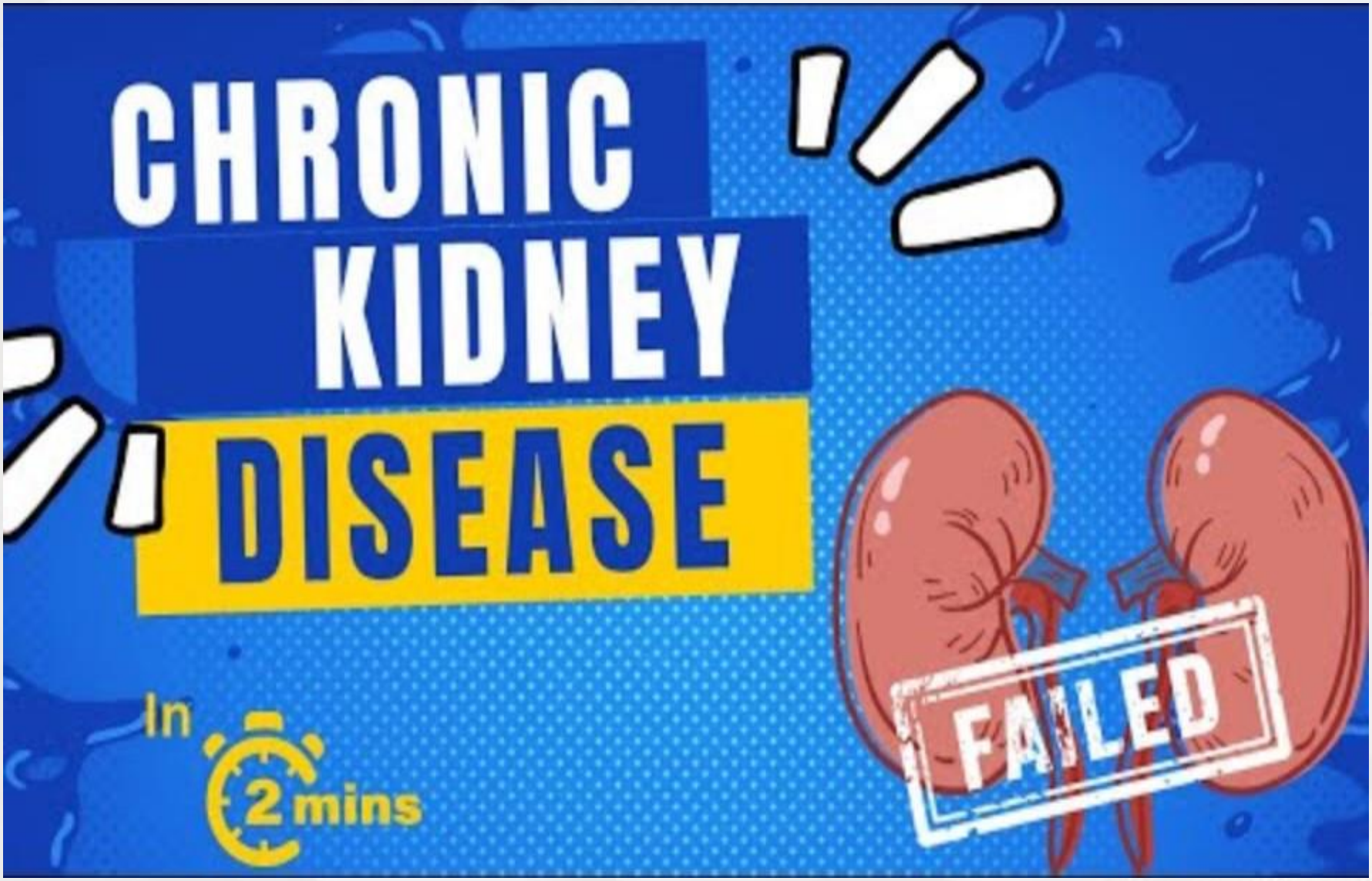
Chronic Kidney Disease



- Each kidney has ~ 1 million nephrons (filtering units)
  - Each nephron includes a glomerulus (filter) and a tubule

Chronic Kidney Disease

- Affects almost every organ and major function in the body
- S&S;
  - Lethargy
  - GI symptoms (anorexia, N/V, thirst)
  - Altered Mental Status
  - Pale, cool, diaphoretic skin, edema
  - Anemia
  - Uremia urea in blood, can cause ureic frost
  - Jaundice (retention of urea, metabolic waste)
  - *Dysrhythmias*
  - [(79) Chronic Kidney Disease | Causes and Clinical Manifestations - in 2 mins! - YouTube]


CHRONIC
KIDNEY
DISEASE
In
2 mins
FAILED

Chronic Kidney disease

- Will likely have associated cardiovascular disease
  - HTN (will develop in most patients if not already present)
  - CAD
  - DM

Chronic Kidney Disease

- Management
  - Supportive (ABC's)
  - Fluid bolus if hypotension present (with caution**)
  - ACP – bicarbonate, vasopressor, calcium carbonate
  - Transport to facility with dialysis capability where available
  - In-patient, outpatient and at home dialysis for long term management

calcum carbonate in addition to salbutamol is best for treating this. You can patch to give salbutamol.

Patients with kidney failiure and CVD are at risk for hypercholemia. That causes reduced forced of contraction, and prolongs repolarization (Widened qrs, eventual sine wave)

Hemodialysis

- therapy which eliminates toxic wastes from the body when the kidney fails
- Blood is removed from the body and filtered through a dialyzer that cleans waste and removes extra fluid – it then returns to the body
  - Access through AV access (artery and vein are connected in surgery
    - Fistula
    - Graft

The Basics of AV Access

Hemodialysis – in clinic/hospital

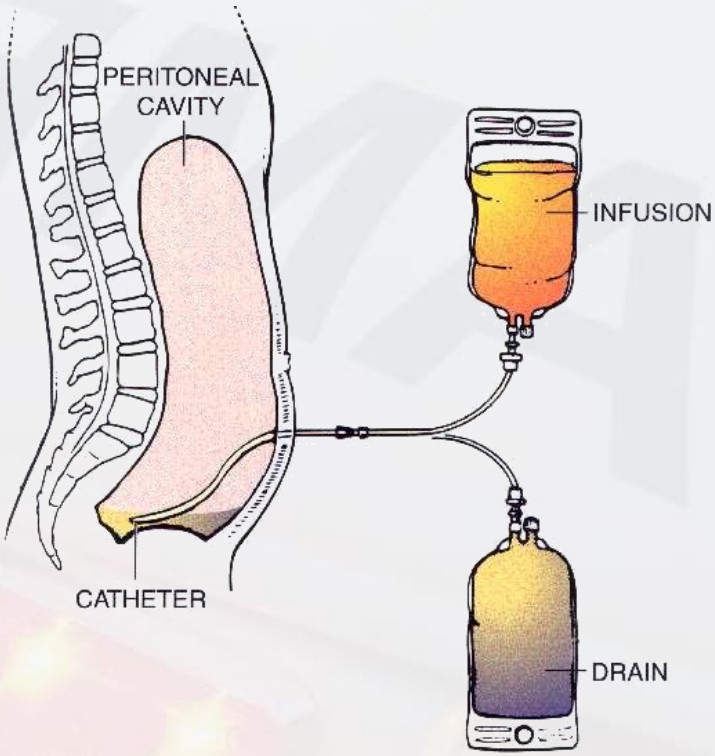
- is performed in a dialysis center for ~ 4 hours three times a week. (every 2-3 days)
- Dialysis is performed through an arteriovenous access (graft/fistula)
- Advantages: No patient training is required.
- Disadvantages: Dialysis graft failure, lack of freedom

Hemodialysis – at home

- Home dialysis machines offer same treatment
- Can be similar time/length or be more often for less time (i.e 2 hours 5-7x/wk)
- Newer technology and machines allowing for more patients to choose at home option
- Requires extensive training for patient and care person

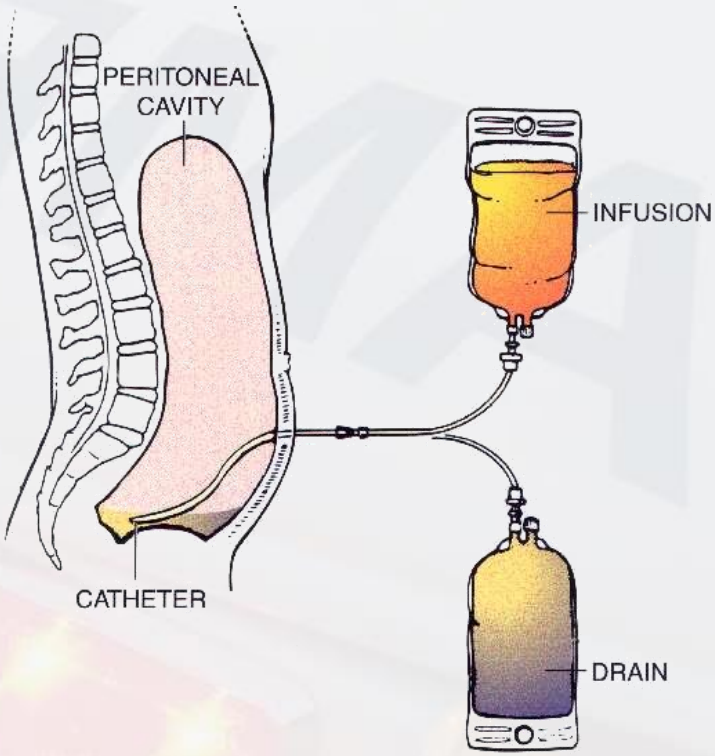
Peritoneal Dialysis

- In peritoneal dialysis the bodies own membrane is used as a filter, and the fluid drained in and out of the abdomen replaces the kidneys in getting rid of the body poisons
  - Is a more continuous form than hemodialysis
  - Requires a less restricted diet
  - done at home, but requires careful technique
  - Allows for more freedom and fewer needle sticks
  - Disadvantages: peritonitis and membrane failure



Peritoneal Dialysis

- Some peritoneal dialysis solutions will alter the blood sugar readings in diabetics (falsely high)
- You may be required to treat with symptom relief based on clinical presentation
  - See local Medical Directives



At Home Dialysis - Disconnect

- May be required in order to transport a critically ill patient and their trained caregiver is unable to do so.
- See link to procedure on Blackboard

QUESTIONS?